

Software Design Description

1. Introduction

This Software Design Description (SDD) presents the design of the Digital Examination Attendance Verification System. The document describes the architecture, components, interfaces, database structure and design decisions used to develop the system.

The system aims to replace the current paper-based examination card process with a digital solution that improves attendance verification, identity verification, examination script submission tracking and record management.

2. Purpose

The purpose of this document is to describe how the proposed system will be designed and implemented. This document serves as a communication tool between the developers, project supervisors, testers, system administrators and future maintenance personnel. It also provides a blueprint that guides software construction and testing.

3. Scope

The digital Examination Attendance Verification System will:

- Verify student identity during examinations.
- Record examination attendance electronically.
- Confirm examination script submission
- Store attendance records securely.
- Generated attendance report
- Maintain audit logs for accountability

The system will not:

- Mark examination scripts
- Manage student grades
- Replace the university registration system

4. References

References

(IEEE), I. o. (2009). *IEEE Standard for Information Technology Systems Design Software Design Descriptions*. New York: IEEE Standards Association.

Sommerville, I. (2016). *Software Engineering 10th edition*. New York: Pearson.

5. System overview

The Existing System - the current system uses physical examination cards. The problems that comes with that include the loss of examination cards, weak student identity verification, manual attendance counting errors, lack of proof of script submission and limited traceability and accountability.

The Proposed System- it is a web-based digital attendance verification system. It will allow for student verification, attendance recording, script submission confirmation, report generation and record tracking.

6. Design Stakeholders and Concerns

Stakeholders include:

- Students - fast attendance verification, Accurate attendance records, and Protection of personal information.
- Invigilators - easy attendance recording, reliable student verification, and accurate attendance reports
- Lecturers - access to attendance records, and reliable reporting
- System administrators - user management, system security, and database maintenance
- University management - accountability, examination integrity, and reliable records

7. Design Viewpoint

The system is designed as a web-based Digital Examination Attendance and Script Verification System. The design focuses on improving the way examination attendance, student verification, seating arrangements, and script submission are managed during university examinations.

The system uses a centralised approach where the scanner is not tied to only one paper. Instead, the central door scanner identifies a student using the scanned student card ID, checks the current date and time, checks the active exam timetable, verifies the student's registration, and then updates the correct exam dashboard. This design is important because a student may be registered for many different papers, but at a particular time the system must know which paper the student is currently writing.

The system also supports large venues where students from different courses may write exams at the same time. For example, students writing CSC322, MAT321, and PAD345 may all be seated in the Main Hall during the same time block. To support this, the system groups exam sessions by venue, date, start time, and end time. It then generates one combined seating plan for that venue and time block. Students are grouped by course and assigned seat numbers in sequence. This allows invigilators to download the seating plan before the exam and place personalised scripts on the correct desks.

- The system separates the following concepts:
- Student card ID identifies the student during scanning.
- Student number identifies the student officially in university records.
- Seat number identifies where the student sits for a particular exam sitting.
- Script code identifies the personalised answer script.
- Exam session identifies a specific paper being written at a specific date, time, and venue.
- Venue exam block identifies all papers written in the same venue at the same date and time.

This separation makes the design more reliable because a seat number can change from one exam to another, while the student card ID remains the same.

The system follows a modular design. Each major function is separated into a clear module. The main modules are:

- User authentication and access control
- Student, course, and department management
- Exam session and registration management
- Venue seating plan management
- Central student entry scanning
- Central script submission scanning
- Dashboard and monitoring
- Report and audit export
- Flagged attempt management

The system uses a database-driven design. All important information, such as student records, exam sessions, seat allocations, attendance records, script records, and flagged attempts, is stored in the database. This allows the system to provide real-time dashboards and generate reports based on stored evidence.

The system is designed to reduce manual errors, prevent duplicate records, detect suspicious cases, and provide reliable evidence when there are disputes about attendance or submitted scripts.

8. Architectural Design

The system will use a Three-Tier Architecture

Presentation Layer- provides user interfaces. e.g. login screen, attendance screen, report screen

Business logic layer. e.g. verify student, record attendance, generate reports and confirm script submission.

Data layer- stores information in a database. E.g. student table, examination table, attendance table and user table

9. Detailed Design

9.1 Authentication and User Access Module

This module controls access to the system. Only authorised users such as administrators, lecturers, examiners, and invigilators are allowed to use the system. Users must log in before accessing the examination control centre, scanners, dashboards, seating plans, and reports.

The authentication module protects sensitive student and examination information from unauthorised access.

9.2 Examination Control Centre Module

The examination control centre acts as the landing page of the system. It provides quick access to the main system functions.

The control centre provides:

- Link to the central student entry scanner
- Link to the central script submission scanner
- List of today's exam sessions
- Summary statistics for expected students, present students, and submitted scripts
- Department browsing
- Course search
- Student search

The purpose of this module is to reduce the time needed to locate a paper or student record.

9.3 Department and Course Management Module

This module allows the system to organise papers by department and course. Each course belongs to a department. Users can browse departments and view all exam sessions under a selected department.

This helps lecturers and examiners find the correct exam dashboard without searching through unrelated papers.

9.4 Exam Session Management Module

An exam session represents one paper written on a specific date, at a specific time, and in a specific venue.

Each exam session stores:

- Course
- Venue
- Exam date
- Start time
- End time

The system uses the exam session information to determine which paper is active when a student scans at the door.

9.5 Exam Registration Module

This module links students to the exam sessions they are allowed to write. A student may be registered for many exam sessions, and each exam session may have many registered students.

When a student scans at the door, the system checks whether the student is registered for the detected active exam session. If the student is not registered, access is denied and the attempt may be flagged.

9.6 Venue Exam Block Module

This module groups all exam sessions that are written in the same venue at the same date and time.

For example, if CSC322, MAT321, and PAD345 are all written in Main Hall from 09:00 to 12:00, the system treats them as one venue exam block.

The venue exam block is used when generating a combined seating plan for the venue. This is important because large venues may contain students from different papers at the same time.

9.7 Seat Allocation Module

This module assigns seat numbers to students for a specific venue exam block.

The system generates seats by grouping students according to course. For example CSC322 students may receive seats 001 to 050, MAT321 students may receive seats 051 to 080, PAD345 students may receive seats 081 to 120.

Each seat allocation links to student, exam session, venue exam block, seat number

The system allows the seating plan to be viewed and exported before the exam. Invigilators can use the exported seating plan to place personalised scripts on the correct desks.

9.8 Central Student Entry Scanner Module

This module is used at the door of the exam venue. The invigilator scans or enters the student card ID.

The system then performs the following steps:

1. Finds the student using the card ID.
2. Checks the current date and time.
3. Finds active exam sessions within the allowed scanning time.
4. Checks whether the student is registered for one of the active exam sessions.
5. Finds the student's seat allocation.
6. Displays the student's details, paper, venue, and seat number.
7. Allows the invigilator to confirm the student's identity.
8. Records the student as present.

If the student is not found, not registered, outside the scan time, or already scanned, the system displays an error or warning message.

9.9 Script Submission Scanner Module

This module is used when students submit personalised answer scripts.

Each script has a unique script code. When the script is scanned, the system identifies:

- The student
- The exam session
- The script status

- The seat number
- Whether the student was recorded at the door

The system then allows the invigilator to confirm the script submission.

If the student was not recorded at the door, the script can still be received but it is marked as flagged. This helps identify cases where a script was submitted without a matching attendance record.

9.10 Dashboard Module

Each exam session has a dashboard that shows the status of the paper.

The dashboard displays:

- Total expected students
- Total present students
- Total missing students
- Total submitted scripts
- Students present but script not submitted
- Scripts submitted without door scan
- Flagged attempts
- Seat information where available

The dashboard helps invigilators and examiners monitor the exam in real time.

9.11 Report and Audit Module

This module allows users to export reports for evidence and record keeping.

The system supports:

- Exam audit report
- Venue seating plan export
- Script status report
- Attendance status report

The reports are exported as CSV files so they can be opened in spreadsheet software.

9.12 Flagged Attempt Module

This module records suspicious or invalid actions.

Examples of flagged attempts include:

- Unknown student card
- Student not registered for the active exam
- Duplicate door scan
- Duplicate script scan
- Script submitted without door scan
- Invalid script code

The flagged attempt module supports accountability and helps examiners investigate irregular cases.

10. Database Design

The system uses a relational database design. The database stores all information required for student verification, exam registration, attendance tracking, seating allocation, script submission, and reporting.

10.1 Student Table

The Student table stores student information.

Main fields include:

- student_number
- full_name
- card_id
- programme
- photo

The student number is the official university student identifier. The card ID is the value scanned from the student card or QR code. The photo is optional but useful for identity verification.

10.2 Department Table

The Department table stores university departments.

Main fields include:

- name
- code

Each department can have many courses.

10.3 Course Table

The Course table stores examination papers or courses.

Main fields include:

- department
- code
- name

Each course belongs to one department.

10.4 ExamSession Table

The ExamSession table stores details of a specific paper being written.

Main fields include:

- course
- venue
- exam_date
- start_time
- end_time

Each exam session belongs to one course. The system uses this table to determine which paper is active at a given time.

10.5 ExamRegistration Table

The ExamRegistration table links students to exam sessions.

Main fields include:

- student
- exam_session

This table ensures that a student can only be marked present for an exam if the student is registered for that exam session.

10.6 VenueExamBlock Table

The VenueExamBlock table stores venue and time block information.

Main fields include:

- venue
- exam_date
- start_time
- end_time

This table groups all papers written in the same venue at the same date and time. It is used when generating a combined seating plan.

10.7 SeatAllocation Table

The SeatAllocation table stores seat numbers assigned to students.

Main fields include:

- venue_block
- exam_session
- student
- seat_number
- assigned_at

Each student can have one seat allocation per exam session. Seat numbers are unique within a venue exam block.

10.8 AttendanceRecord Table

The AttendanceRecord table stores students who were confirmed at the door.

Main fields include:

- student
- exam_session
- scanned_at
- verified_by

This table prevents duplicate attendance records by ensuring that a student is only recorded once for the same exam session.

10.9 ExamScript Table

The ExamScript table stores personalised script records.

Main fields include:

- student
- exam_session
- script_code
- status
- submitted_at
- received_by
- flag_reason

The script code uniquely identifies each personalised answer script. The script status may be Not Submitted, Submitted, or Flagged.

10.10 FlaggedAttempt Table

The FlaggedAttempt table stores suspicious or invalid actions.

Main fields include:

- card_id
- exam_session
- reason
- flag_type
- attempted_at

This table supports auditing by keeping a record of invalid scans, duplicate attempts, and other irregular cases.

11. User Interface Design

11.1 General Interface Design

The system user interface is designed to be simple, clear, and suitable for use during examinations. Since invigilators need to work quickly, the system uses large buttons, clear labels, tables, status badges, and confirmation messages. The interface is web-based and can be accessed using a desktop computer, laptop, or tablet.

11.2 Login Page

The login page allows authorised users to sign in before accessing the system. This prevents unauthorised users from viewing or changing examination data.

11.3 Examination Control Centre

The control centre is the main landing page after login. It provides quick access to the most important system functions.

The page includes:

- Student Entry Scanner button
- Script Submission Scanner button
- Today's exams
- Course search
- Student search
- Department cards
- Summary statistics

This page helps users avoid manually searching through many departments and papers.

11.4 Course Search Interface

The course search interface allows a lecturer or examiner to search for a paper using a course code or course name.

The result shows:

- Course code
- Course name
- Department
- Venue
- Date
- Time
- Dashboard link
- Scanner links

11.5 Student Search Interface

The student search interface allows an authorised user to search for a student using the student number, card ID, or name.

The result shows:

- Student name
- Student number
- Programme
- Registered exam sessions
- Attendance status
- Script status
- Dashboard link

This interface is useful when handling student queries or disputes.

11.6 Central Student Entry Scanner Interface

The central student scanner is used at the exam venue entrance.

The user enters or scans the student card ID. The system then displays:

- Student name
- Student number
- Programme
- Photo if available
- Detected paper
- Venue
- Time
- Seat number
- Confirmation button

The invigilator must confirm that the student details match the person at the door before the student is marked present.

11.7 Central Script Submission Scanner Interface

The central script scanner is used when students submit scripts.

The user scans or enters the script code. The system then displays:

- Student name
- Student number
- Paper
- Venue
- Seat number
- Script code
- Door scan status
- Confirmation button

If there is no door scan record, the system warns the user and marks the script as flagged if confirmed.

11.8 Exam Dashboard Interface

The exam dashboard shows the status of a specific exam session.

It includes:

- Expected students
- Present students
- Missing students
- Submitted scripts
- Present students without submitted scripts
- Scripts submitted without door scan
- Flagged attempts
- Export audit report button
- Seating plan buttons

The dashboard gives examiners and invigilators a clear view of what is happening during and after the exam.

11.9 Venue Seating Plan Interface

The venue seating plan interface shows all students writing in the same venue at the same time.

It displays:

- Seat number
- Course code
- Student number
- Student name
- Programme
- Script code

The seating plan can be exported as a CSV file and printed before the exam. Invigilators use this to place personalised scripts on desks before students enter.

11.10 Script Records Interface

The script records page shows all personalised script records for an exam session.

It displays:

- Student number

- Student name
- Script code
- Script status
- Submission time

This page helps invigilators and examiners track whether each expected script was submitted.

11.11 Report Export Interface

The system provides buttons for exporting audit reports and seating plans. The exported files are in CSV format and can be opened using spreadsheet software.

The reports support evidence keeping and help resolve disputes about attendance or missing scripts.

12.Security design

The system shall provide:

Authentication- only registered users can access the system

Authorization- difference users have different permissions

Administrator- has full access

Invigilator- attendance and submission management

Lecturer- view reports only

Password protection- passwords will be encrypted before storage

Audit logging- all critical activities will be logged

13.Design constraints

The constraints of the design are availability of internet connection, existing university infrastructure, institutional examination policies, database storage capacity

14.Assumptions and dependencies

Assumptions- students are registered before the examination, examination schedules are available, invigilators are trained.

Dependencies- student database, network connectivity, and server infrastructure.

15.Design rationale

The reasons for the selected design is because of the three-tier architecture improves maintainability, digital records reduce loss of examination information, modular design improves scalability, authentication improves security and automated reporting improves efficiency. Based on (Sommerville, 2016) the design follows object-oriented principles and modern software engineering practices.

16.Future enhancements

Future versions may include:

- QR Code verification
- Biometric verification
- Facial recognition
- Mobile application support
- Real time analytics dashboards
- Integration with the university student portals

17.Conclusion

The Digital Examination Attendance Verification system is designed to improve examination attendance management by replacing paper-based processes with a secure digital platform. The design provides an improved attendance verification, better student identity verification, script submission tracking, reliable reporting, and enhanced accountability and traceability. The system architecture and design support maintainability, scalability, security, and future expansion.